

B1 Assignment : Straight line & Circle

SINGLE CORRECT TYPE QUESTIONS

C01/1/001. The point $P(\alpha, \alpha + 1)$ will lie inside the triangle with vertices $A(0, 3)$, $B(-2, 0)$ and $C(6, 1)$ if :

(A) $\alpha = -1$

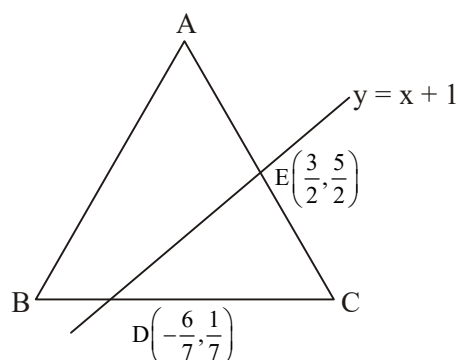
(B) $\alpha = 2$

(C) $-\frac{3}{4} < \alpha < \frac{2}{7}$

(D) $-\frac{6}{7} < \alpha < \frac{3}{2}$

Ans. D

Sol. $-\frac{6}{7} < \alpha < \frac{3}{2}$



C01/1/002. A line passing through point $(2, 2)$ encloses an area of magnitude λ with the axes (not in first quadrant). The intercepts made by the line on the axes are roots of equation :

(A) $x^2 - \lambda x + 2\lambda = 0$

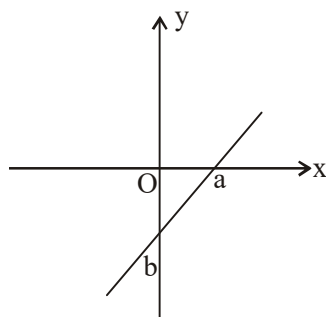
(B) $x^2 + \lambda x - 2\lambda = 0$

(C) $x^2 + \lambda x + 2\lambda = 0$

(D) $x^2 + \lambda x + \lambda = 0$

Ans. B

Sol. Line is $\frac{x}{a} + \frac{y}{b} = 1 \Rightarrow \frac{2}{a} + \frac{2}{b} = 1$ and $\frac{-1}{2}ab = \lambda$



$$\Rightarrow \frac{2}{a} - \frac{a}{\lambda} = 1 \Rightarrow 2\lambda - a^2 = a\lambda \Rightarrow a^2 + \lambda a - 2\lambda = 0$$

$\Rightarrow a, b$ are roots of equation $x^2 + \lambda x - 2\lambda = 0$

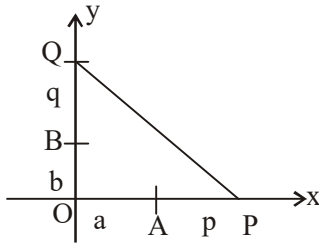
C01/1/003. Let A and B are two points on positive x-axis and positive y-axis respectively, O being origin. If sides OA and OB, each equal to 'a' are produced to P and Q respectively such that AP. BQ = OA . OB then the line PQ always passes through the fixed point :

- (A) (a, a) (B) $\left(\frac{a}{3}, \frac{a}{3}\right)$ (C) $\left(\frac{a}{2}, \frac{a}{2}\right)$ (D) (2a, 2a)

Ans. A

Sol. Given $qp = a^2$

Equation of PQ is



$$\frac{x}{a+p} + \frac{y}{a+q} = 1 \Rightarrow (a+q)x + (a+p)y = 2a^2 + (p+q)a$$

Which passes through (a, a)

C01/1/004. An insect is resting on the graph paper at a point A(3, 2). Now it starts moving towards west direction and covers a distance of 4 units and then it turns towards south and covered a distance of 3 units and reaches at point B then the polar co-ordinates of point B will be :

- (A) $\left(6\sqrt{2}, \frac{\pi}{4}\right)$ (B) $\left(\sqrt{2}, \frac{3\pi}{4}\right)$ (C) $\left(\sqrt{2}, \frac{-3\pi}{4}\right)$ (D) None of these

Ans. C

Sol.

C01/1/005. Let A $\equiv$ (2, 0) and B $\equiv$ (3, 1). The line AB turns about A through an angle $\frac{\pi}{12}$ in the clockwise sense, and the new position of B is B'. Then B' has the co-ordinates :

- (A) $\left(\frac{2\sqrt{2}-\sqrt{3}}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ (B) $\left(\frac{2\sqrt{2}+\sqrt{3}}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
 (C) $\left(\frac{\sqrt{3}-2\sqrt{2}}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ (D) $\left(\frac{\sqrt{3}-2\sqrt{2}}{2}, \frac{1}{\sqrt{2}}\right)$

Ans. B

Sol.

C01/1/006. If perpendicular distance from the point (3,1) to the line $(2a + b)x + (a + b)y - b = 0$ (where $a, b \neq 0$) is maximum then $\left(\frac{b-a}{b+a}\right)$ is equal to :

- (A) 9 (B) 10 (C) 11 (D) 12

Ans. C

Sol. $a(2x + y) + b(x + y - 1) = 0$

above line is a family of

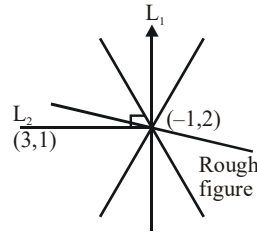
lines passing through

intersection of

$$2x + y = 0 \text{ \& } x + y - 1 = 0$$

$\Rightarrow$ point of intersection is $(-1, 2)$

for maximum distance $(-1, 2)$ must be foot of perpendicular.



$$\therefore m_{L_2} = \left(\frac{2-1}{-1-3}\right) = -\frac{1}{4}$$

$$\therefore m_{L_1} = 4$$

equation of L_1 is $y - 2 = 4(x + 1)$

$$\Rightarrow 4x - y = -6$$

(from compare $a = 5$ & $b = -6$)

$$\therefore \frac{b-a}{b+a} = 11$$

C01/1/007. In $\triangle ABC$, perpendicular BE is drawn from vertex B on internal angle bisector from vertex A such that it divide $\angle B$ in $1 : 2$ (point E lies on internal angle bisector from vertex A and lies inside $\triangle ABC$) and $\triangle ABE$ is isosceles, then value of $\angle C$ will be :

- (A) $\frac{\pi}{8}$ (B) $\frac{3\pi}{8}$ (C) $\frac{\pi}{12}$ (D) $\frac{5\pi}{12}$

Ans. A

Sol. As $AE = BE$

So $\angle ABE = \angle BAE$

$$\Rightarrow \frac{A}{2} = \frac{2B}{3} = \frac{\pi}{4}$$

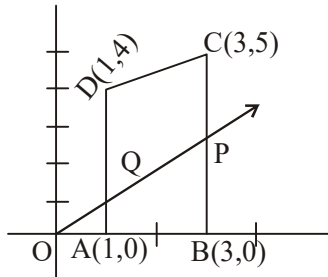
$$A = \frac{\pi}{2}, B = \frac{3\pi}{8} \quad \text{So } \angle C = \frac{\pi}{8}$$

C01/1/008. (1,0),(3,0),(3,5) and (1,4) are the vertices of a quadrilateral. The equation of line through the origin which divides the quadrilateral into two parts of equal area is :

- (A) $y = \frac{9}{8}x$ (B) $y = \frac{8}{9}x$ (C) $y = 8x$ (D) $y = \frac{x}{8}$

Ans. A

Sol.



$$\text{Area of trap} = \frac{1}{2} [4 + 5] \times 2 = 9$$

$$\therefore \text{Area of trap ABPQ} = \frac{9}{2}$$

Let $y = mx$ be equation of OP

Q(1, m) and P(3, 3m)

$$\text{Area ABPQ} = \frac{1}{2} (m + 3m) \times 2 = 4m$$

$$\therefore 4m = \frac{9}{2}$$

$$m = \frac{9}{8}$$

$$y = \frac{9}{8}x$$

C01/1/009. If the line $y = \sqrt{3}x$ cuts the curve $x^4 + ax^2y + bxy + cx + dy + 6 = 0$ at A, B, C and D, then value of OA . OB . OC . OD is, (where O is origin) :

- (A) $a + b + c$ (B) $2c^2d$ (C) 96 (D) 6

Ans. C

Sol.

$$\because |z_1| = 1 \Rightarrow \bar{z}_1 = \frac{1}{z_1} \Rightarrow \bar{z}_1 z_3 - z_4 = \frac{z_3 - z_4 z_1}{z_1}$$

$$\therefore z_2 = \frac{z_1 \cdot (z_3 - z_4)}{(\bar{z}_1 z_4 - 1)} \Rightarrow |z_2| = \frac{|z_3 - z_4|}{|\bar{z}_1 z_4 - 1|} \leq 1$$

$$\Rightarrow |z_3 - z_4|^2 \leq |\bar{z}_3 z_4 - 1|^2$$

$$(z_3 - z_4)(\bar{z}_3 - \bar{z}_4) \leq (\bar{z}_3 z_4 - 1)(z_3 \bar{z}_4 - 1)$$

$$|z_3|^2 + |z_4|^2 - |z_3|^2 |z_4|^2 - 1 \leq 0$$

$$\underbrace{(|z_3|^2 - 1)}_{(-)ve} (1 - |z_4|^2) \leq 0$$

$$\Rightarrow (1 - |z_4|^2) \geq 0 \Rightarrow |z_4| \leq 1$$

C01/1/010. P and Q are two points on the base AB of a triangle ABC whose vertices are $(-2, 3)$, $(4, -6)$ and $(1, 1)$ respectively. If the join of CP and CQ divides the triangle ABC into three triangles of equal areas, then the equation of line pair through the origin and parallel to CP and CQ is equal to :

(A) $y^2 + 4xy - 3x^2 = 0$

(B) $3y^2 + 4xy + x^2 = 0$

(C) $y^2 + 3xy - 4x^2 = 0$

(D) $y^2 - 5xy + 4x^2 = 0$

Ans.

C

Sol.

$$AB \rightarrow 3x + 2y = 0$$

$$\text{Let } P\left(h, -\frac{3}{2}h\right)$$

$$\text{and } Q\left(k, \frac{-3k}{2}\right)$$

$$\Delta = \frac{1}{2} \begin{vmatrix} -2 & 3 & 1 \\ 4 & -6 & 1 \\ 1 & 1 & 1 \end{vmatrix} = \frac{15}{2}$$

$$\frac{1}{2} \begin{vmatrix} h & -3/2h & 1 \\ 1 & 1 & 1 \\ -2 & 3 & 1 \end{vmatrix} = \frac{1}{3} \times \frac{15}{2} \Rightarrow h = 0$$

$$\Rightarrow P \equiv (0, 0)$$

$$\frac{1}{2} \begin{vmatrix} k & -3/2k & 1 \\ 1 & 1 & 1 \\ -2 & 3 & 1 \end{vmatrix} = \frac{2}{3} \times \frac{15}{2} \Rightarrow k = 2$$

$$\Rightarrow Q \equiv (2, -3)$$

$$M_{PC} = 1 \quad M_{QC} = 4$$

$$(y - x)(y + 4x) \equiv y^2 + 3xy - 4x^2 = 0$$

